

EVALUATING REAL-WORLD PROBLEMS USING THE ORDER OF OPERATIONS – PART 2



LESSON INTRODUCTION

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers, including exponents and radicals.

LEARNING TARGETS

- I can solve real-world problems involving the order of operations with rational numbers, including exponents and radicals.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do I solve real-world problems involving the order of operations?

LESSON NARRATIVE

In this final lesson of Unit 12 students will work through three activity stations. At these stations, students will analyze and evaluate expressions that model real-world contexts.

COMPONENT SUMMARY

12.10.1 WARM-UP (5 MIN.)

Find the value of an expression

12.10.3 ACTIVITY: STATION 2: (12 MIN.)

Evaluate expressions based on generating power

12.10.5 WRAP-UP (~5 MIN.)

Reflect on the lesson

12.10.2 ACTIVITY: STATION 1: (15-17 MIN.)

Analyze and evaluate an expression based on home renovations

12.10.4 ACTIVITY: STATION 3: (14-17 MIN.)

Evaluate an expression and an equation based on planting strawberries

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- (Optional) Four Function Calculators

ADDITIONAL PREPARATION

- Consider strategic grouping prior to the start of the lesson for 12.10.2, 12.10.3, and 12.10.4 Activities.
- Set up the stations prior to the start of the lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may apply the incorrect order of operations by incorrectly applying the Laws of Exponents, or by not recognizing values that are enclosed by grouping symbols that need to be evaluated first, or by assuming they should evaluate from left to right regardless of the operation.

THINGS TO CONSIDER

- Allowing students to use calculators during the lesson will help students focus on discovery and process of evaluating real-world problems. Consider having non-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with evaluating. For example, question 1 in 12.10.1 Warm-Up could be a non-calculator problem but question 2 could be a calculator problem.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

During the 12.10.2 Activity, students will relate the expression to the information in the table. Consider incorporating this routine. After providing students think time, pair the partners with another pair and have the pairs share their responses with each other. The Think-Pair-Share routine can be utilized throughout 12.10.2, 12.10.3, and 12.10.4 Activities.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

12.10.2, 12.10.3, and 12.10.4 Activities are grounded in students solving real-world problems involving the order of operations. At each station, students connect the mathematical concepts explored in Unit 12 to real-world contexts. Students will validate conclusions by comparing them with their partner (or group).

DIFFERENTIATE INSTRUCTION

12.10.2, 12.10.3, and 12.10.4 Activities consist of three stations at which students are applying what they have learned about evaluating expressions using the order of operations to real-world contexts. Consider a variety of strategies for engaging students with the content.

- Each small group completes each of the three stations. For small groups of 3-4 students, consider having 3 copies of each station for a total of 9 groups that can work simultaneously. After all the groups have completed all three stations, have one student from each group pair up with a different group to share responses to verify accuracy and compare strategies.
- Each small group is assigned one station at which they are experts and create an anchor chart to display their work. After each group completes their work, every student then pairs with a student from another group and both students act as experts or teachers of their stations. Students repeat this process for the third station.

The class can also be broken into 6 small groups and there can be two sets of each station. Students can either rotate through the stations or the stations can be passed from group to group.

12.10.1 WARM-UP: BELL WORK

TEACHER MOVES

Use student understanding to inform your instruction and to determine where during this and further lessons. Make a list of misconceptions and determine which should be addressed whole-group, and which should be addressed in a small group with those who are struggling to be proficient.



12.10.1 Warm-Up: Bell Work

Find the value of each expression. Show your work.

1. $(3 \cdot \frac{2}{3})^{-2} - 4(6 + \frac{7}{2})$

$$(2)^{-2} - 4(6 + \frac{7}{2})$$

$$(2)^{-2} - 4(\frac{12}{2} + \frac{7}{2})$$

$$(2)^{-2} - 4(\frac{19}{2})$$

$$\frac{1}{2^2} - 4(\frac{19}{2})$$

$$\frac{1}{4} - 4(\frac{19}{2})$$

$$\frac{1}{4} - \frac{76}{2}$$

$$\frac{1}{4} - \frac{152}{4}$$

$$-\frac{151}{4} = -37\frac{3}{4}$$

2. $\frac{5.5^2 - 9.25}{7} - \sqrt{11 - 2}$

$$\frac{30.25 - 9.25}{7} - \sqrt{11 - 2}$$

$$\frac{30.25 - 9.25}{7} - \sqrt{9}$$

$$\frac{30.25 - 9.25}{7} - 3$$

$$\frac{21}{7} - 3$$

$$3 - 3$$

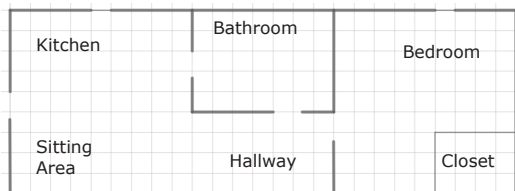
$$0$$



12.10.2 Activity: Station 1: Home Renovations

Liliana is looking to do some home renovations on her apartment. The floor plan of her apartment is shown, where each gridline measures 1 foot (ft). She plans to re-tile the floor of the bathroom and lay new carpeting on the floor of the bedroom, including the closet.

Her bedroom, including the closet, measures 9 ft by 9 ft, and her bathroom is 5 ft by 7 ft.



The prices Liliana found for the materials she needs to update the flooring in both rooms is given in the table.

	Bedroom			Bathroom	
	Carpet	Padding Material	Underlay	Tile	Grout
	Home Decorators Sycamore	Memory Foam	OSA	Marazzi	Polyblend
Price	\$2.89/ft ²	\$0.75/ft ²	\$1.00/ft ²	\$1.89/ft ²	\$13.48

12.10.2 ACTIVITY: STATION 1: HOME RENOVATIONS

TEACHER MOVES

Facilitation of 12.10.2, 12.10.3, and 12.10.4 Activities:

- The class can be broken into 6 small groups and there can be two sets of each station. Students can either rotate through the stations, or the stations can be passed from group to group.
- Assess the needs of your students to determine if an alternate set up would be more beneficial. • Consider giving students a four-function calculator to use at the stations. While they may not have a calculator on the FSA, they also likely will see simpler real-world contexts. This is a lot to do in one class period by hand, but the contexts are more engaging and realistic.
- Monitor while students are working in each station to determine if there are any questions that need to be debriefed. This could include highlighting student voices to share rich discussions, or comments overhead while they collaborated for the whole class to hear, or to address common misconceptions that were evident among the majority of the class.

ENRICHMENT

Prior to completing question 1, encourage students to record information on the diagram of Liliana's home provided. Ask students questions about what information is important to assist them with their recordings, such as:

- Which areas of Liliana's home could you place an "X" on that are not necessary for 12.10.2 Activity?
- What information should you record in Liliana's bathroom that will help you with 12.10.2 Activity?
- What information should you record in Liliana's bedroom that will help you with 12.10.2 Activity?

- Liliana writes the following expression to determine the total cost of installing flooring in both rooms.

$$9^2(2.89 + 0.75 + 1) + (5 \cdot 7)(1.89) + 13.48$$

- Discuss with your partner how the expression relates to the materials in the table.
- Circle the part of the expression that represents the cost of carpeting the bedroom.
- Underline the part of the expression that represents the cost of tiling the bathroom.
- Evaluate the expression to determine how much it will cost Liliana for the materials to replace the flooring in both rooms. Show your work.

$$\begin{aligned}
 &9^2(2.89 + 0.75 + 1) + (5 \cdot 7)(1.89) + 13.48 \\
 &9^2 \cdot 4.64 + (5 \cdot 7)(1.89) + 13.48 \\
 &9^2 \cdot 4.64 + 35 \cdot 1.89 + 13.48 \\
 &81 \cdot 4.64 + 35 \cdot 1.89 + 13.48 \\
 &375.84 + 35 \cdot 1.89 + 13.48 \\
 &375.84 + 66.15 + 13.48 \\
 &455.47
 \end{aligned}$$

- Liliana is concerned about the total cost and wants to know how much it would cost to re-floor the bedroom only. How much would the flooring materials cost for only her bedroom?

$$\$375.84$$

12.10.3 ACTIVITY: STATION 2: GENERATING POWER

TEACHER MOVES

Consider having students use the Think-Pair-Share routine for both parts of question 1. This will allow students the opportunity to first read through complex text, identify the important information, and then evaluate the expression based on the given context. Students will pair after evaluating each expression to share their answers and discuss their process in evaluating. After students share, they should be prompted to revisit their answers and make any revisions if needed.



12.10.3 Activity: Station 2: Generating Power

1. After several power outages caused by storms, Phillip purchased a used generator for \$1,380 three years ago. Equipment loses value each year from basic wear and tear and as newer models come on the market. Phillip estimates the generator loses about 20% of its value each year based on prices he sees online.
 - a. Phillip uses the expression $1380 \cdot 0.8^3$ to determine a fair price for which to sell the generator now. What is the fair price Phillip determined?
\$706.56
 - b. If Phillip's estimation of how much value the generator loses each year is accurate, he can use the expression $1380 \cdot 0.8^{-2}$ to determine the price of the generator two years before he bought it when it was new. Evaluate the expression to find the price of the generator when it was new.
\$2,156.25



12.10.4 Activity: Station 3: Planting Strawberries

1. Michaela is considering planting strawberries for the first time on her farm. She has an acre set aside and wants to determine whether the costs and potential profits make sense.

After researching, she determines the following information.

Costs of Planting 1 Acre of Strawberries	Expected Revenue from 1 Acre of Strawberries
- Plants: \$1500 - Labor: \$595 - Materials: \$1724	- Total plants: 15,000 - Expected percentage available for sale: 70% - Average revenue per plant: \$1.21

The expression shown can be used to find the amount of money Michaela can expect to earn based on her research.

$$1.21(0.70 \cdot 15,000) - (1500 + 595 + 1724)$$

- a. Discuss with your partner how the expression relates to the information in the table.

12.10.4 ACTIVITY: STATION 3: PLANTING STRAWBERRIES

TEACHER MOVES

Prompt students to discuss with their partner how the expression relates to the table from the context. Urge students to include justifying the use of the operations in the expression instead of only labeling the information that each number relates to in the table. Students may realize that the operation in front of the subtraction symbol is multiplication and not addition, like how the cost is calculated, because the revenue is calculated from an average of a percentage. Students may also point out that subtraction is used because the costs must be subtracted from the revenue.

12.10.4 ACTIVITY: STATION 3: PLANTING STRAWBERRIES (CONTINUED)

TEACHER MOVES

If students are working in pairs, consider having pairs form a group of four to compare responses to question 1c.

- b. Profit can be calculated using the equation shown.

$$\text{Revenue} - \text{Costs} = \text{Profit}$$

Work with your partner to explain how this relates to the expression Michaela wrote.

Answers will vary. Michaela substituted the values for her Revenue, $1.21(0.70 \cdot 15,000)$ and substituted the values for her Costs, $(1500 + 595 + 1724)$. In her expression, she subtracted them to find the Profit.

- c. Determine the profit Michaela is expecting to earn based on her research.

\$8,886



12.10.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target in the table.

Answers will vary.



I need help.



I got it.



I could help someone.

Evaluate expressions with rational numbers, exponents, and radicals using the order of operations.	  
Solve real-world problems by evaluating expressions using the order of operations.	  

12.10.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Use the data from this self-assessment, along with observational data from the lesson, to determine what concepts, if any, need remediation with before the Unit Assessment.